

Peter Bremer-Feit

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EDUCATION

University of Illinois Urbana-Champaign <i>Master of Computer Science</i> <i>Joint BS-MCS Program</i>	Dec 2025 <i>Current GPA: 4.0/4.0</i>
University of Illinois Urbana-Champaign <i>Bachelor of Science in Computer Science</i> <i>Focus Areas: Intelligence and Big Data, Algorithms and Models of Computation</i>	Dec 2024 <i>GPA: 4.0/4.0</i>
Illinois Institute of Technology <i>Dual Enrollment</i>	August 2020 <i>GPA: 4.0/4.0</i>

EXPERIENCE

Capital One <i>Machine Learning & Technology Intern</i>	Summer 2025
- Designed and implemented a decoder only transformer model to decide what messages to show to Capital One app users. - Wrote an efficient pipeline to train the model off of millions of rows of user extensive data. - Worked in Databricks, Snowflake EMP notebooks and PyTorch	
Chamberlain Group <i>Test Automation Software Engineer Intern</i>	Summer 2024
- Designed and wrote a program using LabVIEW, Appium, Python and Arduino to test the person detection and facial recognition capabilities of Chamberlain's Video Keypad. - Constructed a testing rig to simulate moving a mannequin around in 2D space for camera testing. - Wrote a library to interact with a Copley Controls' Motor Drive through LabVIEW. - Developed skills working along side a team using agile planning and scum boards.	

SKILLS

Languages: C/C++, Java, Python, HTML/CSS, SQL, OCaml, Verilog, LaTeX, LabVIEW
Tools/Libraries: Git/GitHub, NumPy, Pandas, Spark, PyTorch, Databricks, Snowflake, Matplotlib, GDB, Valgrind, Bash, pthreads, OpenMP, MPI, Requests, Selenium, BeautifulSoup, pybluez, Jupyter Notebooks, Appium, Arduino, Windows, Linux

PROJECTS

Campus and Course Mapping System <i>Django, SQLite, HTML/CSS, PythonAnywhere, Git</i>	Spring 2023
- Developed a Campus and Course Mapping System at UIUC. - Enabled seamless campus navigation, room selection, and class section information access. - Implemented robust search features for course section filtering. - Utilized web scraping libraries to automatically update course sections.	
Device Location Tracker <i>Python, Raspberry Pi, Bluetooth, pybluez, tkinter</i>	April 2024
- Wrote a program to locate any Bluetooth enabled device within a house. - Made Bluetooth beacons that could be placed around a house to track the location of devices.	
Rubik's Cube Solver <i>Python, NumPy, OpenCV</i>	November 2023
Wrote a program that scans Rubik's Cubes and gives the moves to solve them.	
Image-Based Lighting <i>Python, OpenCV, NumPy, Matplotlib, Jupyter Notebooks, Blender</i>	November 2023
- Reconstructed the lighting in a room from an HDR photo of a spherical mirror. - Convert photo of a spherical mirror into equirectangular domain, allowing for use in blender renders.	
Custom Malloc and Valgrind <i>C, Git</i>	October 2023
- Wrote versions of malloc, calloc, realloc and free in C which performed on par with the official C implementation. - Implemented a simplified version of Valgrind to track memory leaks in C code.	
Gradient Domain Fusion <i>Python, OpenCV, NumPy, Matplotlib, Jupyter Notebooks</i>	October 2023
- Implemented an algorithm able to seamlessly blend images together using Poisson blending. - Automatically shifts image color to better match background image.	
Linux Shell <i>C, ssh, fork, exec, Git</i>	September 2023
Coded a shell in C similar to the Linux shell Bash.	
Image Quilting and Texture Transfer <i>Python, OpenCV, NumPy, Matplotlib, Jupyter Notebooks</i>	September 2023
Developed code for synthesising textures and rendering objects with textures taken from other objects.	